Assignment 4: Linear programming

Question 1 (18 Marks)

For the objective function and constraints shown below, answer the following questions.

i. Generate a composite diagram (i.e., a graphical plot) for the problem. (5 marks)
ii. Identify the feasible solution region in your graphical plot. (2 marks)
iii. Determine the optimum values of x_1, x_2 and z and show where they are located on the graph. (2 marks)
iv. Can the solution space be made unbounded? If yes, explain how. If no, explain why. (5 marks)
v. Show the unbounded solution region. (4 marks)

Maximise: $z = 5x_1 + 9x_2$

Subject to:

$$2x_2 \leq 12$$
$$3x_1 + 4x_2 \leq 32$$
$$4x_1 - 6x_2 \leq 6$$
$$x_1 \geq 0$$
$$x_2 \geq 0$$

Question 2 (20 Marks)

For the objective function and constraints shown below, answer the following questions.

i. Use the automated simplex method to maximize the objective function. (11 marks)

Note: show all arrays and determine the optimum values of x_1, x_2 and z.
Maximise: $z = -4x_1 + 6x_2$

Subject to:

$$4x_2 \geq 8 + x_1$$
$$6 \geq 3x_1 - 3x_2$$
$$32 \geq 2x_1 + 3x_2$$
$$x_1 \geq 0$$

$$x_2 \geq 0$$

ii. Generate a composite diagram (i.e., a graphical plot) for the problem. (3 marks)

iii. Identify the feasible solution region in your graphical plot. (2 marks)

iv. Determine the optimum values of x_1, x_2 and z on the graph and show where they are located. (2 marks)

v. How is the feasible solution region of this problem different from that of Question 1(ii)? (2 marks)

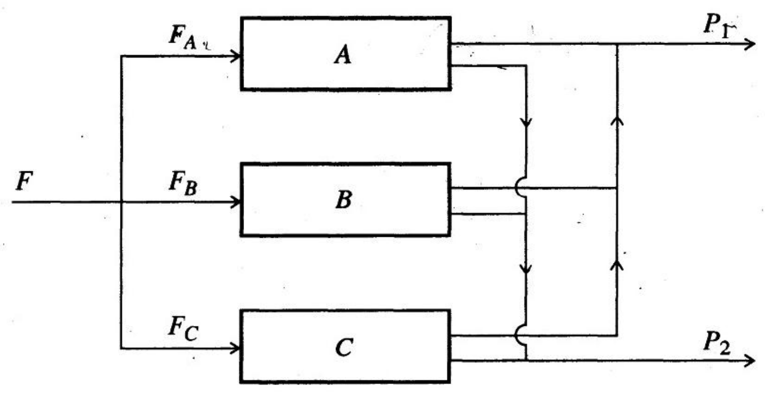
Question 3 (14 Marks)

(EHL, 7.3) Feed to three units (see figure) is split into three streams: F_A, F_B, and F_C. Two products are produced: P_1 and P_2, and the yield in weight percent by unit is

Yield (weight %)	Units A	Unit B	Unit C
P_1	45	60	40
P_2	55	40	60

Each stream has values in $\$/lb$ as follows:

Stream	F	P_1	P_2
Value ($\$/lb$)	0.40	0.56	0.28



Because of capacity limitations, certain constraints exist in the stream flows:

- The total input feed must not exceed $9{,}200\ lb/day$
- The feed to each of the units A, B and C must not exceed $4{,}800\ lb/day$
- No more than $3{,}900\ lb/day$ of P_1 can be used, and no more than $6{,}800\ lb/day$ of P_2 can be used.

In order to determine the values of F_A, F_B and F_C that maximise the daily profit, prepare a mathematical statement of this problem as a linear programming problem and solve to determine the optimal daily profit.

Note: state the optimal values for all variables in the problem.

Question 4 (20 Marks)

(EHL, 7.6): An oil refinery must blend gasoline. Suppose that the refinery wishes to blend four petroleum constituents into three grades of gasoline: A, B, and C. Determine the mix of the four constituents that will maximise profit. The availability and costs of the four constituents are given in the following table:

Constituent *	**Maximum quantity available (barrel/day)**	**Cost per barrel ($)**
1	3100	12.20
2	2100	14.50
3	4100	13.50
4	1300	14.00

*1 = butane, 2 = straight-run, 3 = thermally cracked, 4 = catalytic cracked.

To maintain the required quality for each grade of gasoline, it is necessary to specify certain maximum or minimum percentages of the constituents in each blend. These are shown in the following table, along with the selling price for each grade.

Grade	Specification	Selling price per barrel ($)
A	Not more than 45% of 1	16.50
	Not less than 35% of 2	
	Not more than 50% of 3	
B	Not more than 10% of 1	15.85
	Not less than 10% of 2	
C	Not more than 20% of 1	15.20
	Not more than 20% of 2	
	Not less than 10% of 3	

Assume that all other cash flows are fixed so that the "profit" to be maximized is total sales income minus the total cost of the constituents. Set up a linear programming model for determining the amount and blend of each grade of gasoline. Solve to determine the optimum.

Note: state the optimal values for all variables in the problem.

Question 5 (14 Marks)

(EHL, 7.25): Consider a typical linear programming example in which N grades of paper are produced on a paper machine. Due to raw materials restrictions, not more than a_i tons of grade i can be produced in a week. Let

$x_i =$ numbers of tons of grade i produced during the week.

$b_i =$ number of hours required to produce a ton of grade i

$p_i =$ profit per ton of grade i

Because 165 production hours are available each week, the problem is to find non-negative values of x_i, $i = 1, \dots\dots\dots\dots\dots.., N$, and the integer value N that satisfy

$$x_i \leq a_i$$

$$\sum_{i=1}^{N} b_i x_i \leq c$$

and that maximise the profit function

$$f(x_1, \dots\dots\dots\dots\dots, x_N) = \sum_{i=1}^{N} p_i x_i$$

Data:

	a_i	b_i	p_i
1	410	0.21	20
2	310	0.40	50
3	210	0.20	20
4	110	0.22	13
5	55	0.20	13

$c = 165$

Note: state the optimal values for all variables in the problem.

Question 6 (14 Marks)

(EHL 7.24) Ten grades of crude are available in the quantities shown in the table ranging from 10,000 to 30,000 barrels per day each, with an aggregate availability of 200,000 barrels per day. Refineries X, Y and Z have incremental operations with stated requirements totaling 180,000 barrels per day. Of the available crude, 20,000 barrels per day is not used. One of the refineries can operate at two incremental operations, X_1 and X_2, which represent different efficiency levels. The net profit or loss for each crude in each refinery operation is given on the table in cents per barrel. It is assumed that the crude evaluations reflect the resulting product distribution from these incremental operations. (In practice, however, if further debits are encountered in the solution because of lack of product quality or for transportation of surplus products, suitable corrections can be made in the crude evaluations and the problem reworked until a realistic solution is obtained.)

Maximise the profit per day by allocating the ten crudes among the three refineries with X being able to operate at two levels, so specify X_1 and X_2 as well as Y and Z.

Crude evaluations, availability and requirement

Crude	*a*	*b*	*c*	*d*	*e*	*f*	*g*	*h*	*i*	*j*	Required (M bpd)
Refinery	(Profit or loss of each refinery in cpb)										
X_1	-6	3	17	10	-63	34	15	22	-2	15	**30**
X_2	-11	-7	-16	9	49	16	4	10	-8	8	**40**
Y	-7	3	16	13	-60	25	12	19	4	13	**50**
Z	-1	0	-13	3	48	15	7	17	9	3	**60**
Available (M bpd)	**30**	**30**	**20**	**20**	**10**	**20**	**20**	**10**	**30**	**10**	**200**

Abbreviations: M = 1,000, bpd = barrels per day, cpd = cents per barrel